

New fractional derivative with sigmoid function as the kernel and its models

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ABSTRACT

Based on the idea of the fractional derivative with respect to another function, a new fractional derivative operator with sigmoid function as the kernel in this article, is proposed for the first time. Then, we make use of this new fractional operator to model various nonlinear phenomena from different fields of applications in science, such as the population growth, the shallow water wave phenomena and reaction-diffusion processes, and so on. As a result, we hope that the new fractional operator can be used to discover more evolutionary mechanisms of these phenomena.

1. Introduction

Fractional calculus (FC) has been utilized to model different natural phenomena appear in the areas of application science in recent years [1–3]. For example, water wave movement [4,5], population growth [6], signal processing [7], viscoelastic mechanical evolution mechanisms [8,9], diffusion processes [10–12], reaction-diffusion processes [13,14] and other. We expressed the view that it can be better express nonlinear phenomena than integral order differential model [15,16]. Based on this reason, many scholars proposed disparate definitions of fractional derivatives to deal with this type problem, such as the Riemann-Liouville (RL) fractional derivative [17]

$${}^{RL}D_a^\alpha g(x) = \frac{1}{\Gamma(n-\alpha)} \left(\frac{d}{dx}\right)^n \int_a^x (x-t)^{n-\alpha-1} g(t) dt,$$

the Caputo fractional derivative [18]

$${}^CD_a^\alpha g(x) = \frac{1}{\Gamma(n-\alpha)} \int_a^x (x-t)^{n-\alpha-1} \left(\frac{d}{dx}\right)^n g(t) dt,$$

the Yang et al. RL type fractional derivative [19]

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$${}^{RL}_{E_{\varphi,\phi}}D_a^{(\beta)}g(x) = \frac{d}{dx} \int_a^x E_{\varphi,\phi}((\theta_1, \vartheta_1), (\theta_2, \vartheta_2), \dots, (\theta_n, \vartheta_n); (x-t)^\beta) g(t) dt,$$

the Yang et al. Liouville-Caputo type fractional derivative [19]

$${}^C_{E_{\varphi,\phi}}D_a^{(\beta)}g(x) = \int_a^x E_{\varphi,\phi}((\theta_1, \vartheta_1), (\theta_2, \vartheta_2), \dots, (\theta_n, \vartheta_n); (x-t)^\beta) \frac{d}{dx} g(t) dt,$$

etc [20–22].

In recent years, the RL, Caputo, Hilfer-Katugampola and ψ -Hilfer fractional derivative with respect to another function by some experts, were proposed [2,23–25]. By taking the kernel function as a special function, these differential operators can be reduced into the RL, Caputo, Hilfer, Hilfer-Hadamard, weyl, Chen, Jumarie, Katugampola, Caputo-Hadamard, Prabhakar, Erdelyi-Kober, Riesz, Feller, and Riesz-Caputo fractional derivatives, etc [26–31]. According to this idea, a new fractional differential operator with sigmoid function as the kernel in this article, is proposed. Next, the new fractional derivative operator is applied to write the population growth, the shallow water wave phenomena and reaction-diffusion processes, and so on. Thereby, it can be utilized to explore this more evolutionary mechanisms of these processes.

2. Basic knowledge

In the first place, we give some basic texts on the definitions of fractional integrals and fractional derivatives which will be used to establish this article [2,23,25,32].

Definition 2.1. [2,23,25] The weighted space $C_{\gamma,\psi}[a,b]$ of function f on $(a,b]$ is defined by

$$C_{\gamma,\psi}[a,b] = \{f : (a,b) \rightarrow \mathbb{R}, (\psi(t) - \psi(a))^\gamma f(t) \in C[a,b], 0 \leq \gamma < 1\}$$

with the norm

$$\|f\|_{C_{\gamma,\psi}[a,b]} = \|(\psi(t) - \psi(a))^\gamma f(t)\|_{C[a,b]} = \max_{t \in [a,b]} |(\psi(t) - \psi(a))^\gamma f(t)|. \quad (1)$$

Definition 2.2. [2,23,25] The weighted space $C_{\delta,\psi}^n[a,b]$ of function f on $(a,b]$ is defined by

$$C_{\delta,\psi}^n[a,b] = \{f : (a,b) \rightarrow \mathbb{R}, f(t) \in C^{n-1}[a,b], f^{(n)}(t) \in C_{\gamma,\psi}[a,b], 0 \leq \gamma < 1\}$$

with the norm

$$\|f\|_{C_{\delta,\psi}^n[a,b]} = \sum_{k=0}^{n-1} \|f^{(k)}\|_{C[a,b]} + \|f^{(n)}\|_{C_{\gamma,\psi}[a,b]}. \quad (2)$$

Definition 2.3. (25,32) Take (a,b) ($-\infty \leq a < b \leq +\infty$) be a finite or infinite interval of the real line $\mathbb{R}$ and $\alpha > 0$. Also let $\psi(x)$ be an increasing and positive function monotone function on $(a,b]$, having a continuous derivative $\psi'(x)$ on (a,b) . The left and right-sided fractional integrals of a function f with respect to another function ψ on $[a,b]$ are given by

$$I_{a^+}^{\alpha,\psi} f(x) = \frac{1}{\Gamma(\alpha)} \int_a^x \frac{\psi'(t)}{(\psi(x) - \psi(t))^{1-\alpha}} f(t) dt, \quad (3)$$

and

$$I_{b^-}^{\alpha,\psi} f(x) = \frac{1}{\Gamma(\alpha)} \int_x^b \frac{\psi'(t)}{(\psi(t) - \psi(x))^{1-\alpha}} f(t) dt. \quad (4)$$

Definition 2.4. [2,23,25] Let $\psi'(x) \neq 0$ ($-\infty \leq a < b \leq +\infty$) and $\alpha > 0, n \in \mathbb{N}$. The RL fractional derivative of a function f with respect to $\psi(x)$ of order α correspondent to the RL, are defined by

$$\begin{aligned}
D_{a^+}^{\alpha;\psi} f(x) &= \left(\frac{1}{\psi'(x)} \frac{d}{dx} \right)^n I_{a^+}^{n-\alpha;\psi} f(x) \\
&= \frac{1}{\Gamma(n-\alpha)} \left(\frac{1}{\psi'(x)} \frac{d}{dx} \right)^n \int_a^x \frac{\psi'(t)}{(\psi(x) - \psi(t))^{1+\alpha-n}} f(t) dt,
\end{aligned} \tag{5}$$

and

$$\begin{aligned}
D_{b^-}^{\alpha;\psi} f(x) &= \left(-\frac{1}{\psi'(x)} \frac{d}{dx} \right)^n I_{b^-}^{n-\alpha;\psi} f(x) \\
&= \frac{1}{\Gamma(n-\alpha)} \left(-\frac{1}{\psi'(x)} \frac{d}{dx} \right)^n \int_x^b \frac{\psi'(t)}{(\psi(t) - \psi(x))^{1+\alpha-n}} f(t) dt,
\end{aligned} \tag{6}$$

where $n = [\alpha] + 1$.

Definition 2.5. [25,32] Let $n-1 < \alpha < n$ with $n \in \mathbb{N}$, $I = [a, b]$ ($-\infty < a < b \leq +\infty$) be the interval such that $f(x)$ and $\psi(x) \in C^n([a, b], \mathbb{R})$ two functions such that $\psi(x)$ is an increasing and $\psi'(x) \neq 0$, for all $x \in I$. The ψ -Hilfer fractional derivative (left-sided and right-sided) ${}^H D_{b^-}^{\alpha;\beta;\psi}(\cdot)$ of function of order α and type $0 \leq \beta \leq 1$, are shown by

$${}^H D_{a^+}^{\alpha;\beta;\psi} f(x) = I_{a^+}^{\beta(n-\alpha);\psi} \left(\frac{1}{\psi'(x)} \frac{d}{dx} \right)^n I_{a^+}^{(1-\beta)(n-\alpha);\psi} f(x), \tag{7}$$

and

$${}^H D_{b^-}^{\alpha;\beta;\psi} f(x) = I_{b^-}^{\beta(n-\alpha);\psi} \left(-\frac{1}{\psi'(x)} \frac{d}{dx} \right)^n I_{b^-}^{(1-\beta)(n-\alpha);\psi} f(x). \tag{8}$$

Lemma 1. [2] Let $\alpha > 0$ and $\delta > 0$.

(I) If $f(x) = (\psi(x) - \psi(a))^{\delta-1}$, then

$$I_{a^+}^{\alpha;\psi} f(x) = \frac{\Gamma(\delta)}{\Gamma(\alpha + \delta)} (\psi(x) - \psi(a))^{\alpha+\delta-1}.$$

(II) If $g(x) = (\psi(b) - \psi(x))^{\delta-1}$, then

$$I_{b^-}^{\alpha;\psi} g(x) = \frac{\Gamma(\delta)}{\Gamma(\alpha + \delta)} (\psi(b) - \psi(x))^{\alpha+\delta-1}.$$

Lemma 2. [2] Let $\alpha > 0$ and $\delta > 0$.

(I) If $f(x) = (\psi(x) - \psi(a))^{\delta-1}$, then

$$D_{a^+}^{\alpha;\psi} f(x) = \frac{\Gamma(\delta)}{\Gamma(\delta - \alpha)} (\psi(x) - \psi(a))^{\alpha+\delta-1}.$$

(II) If $g(x) = (\psi(b) - \psi(x))^{\delta-1}$, then

$$D_{b^-}^{\alpha;\psi} g(x) = \frac{\Gamma(\delta)}{\Gamma(\delta - \alpha)} (\psi(b) - \psi(x))^{\alpha+\delta-1}.$$

Remark 1. We note that the definitions (7) and (8) can be rewritten as the following forms with the aid of (3) – (6) and $\chi = \alpha + \beta(n - \alpha)$,

$${}^H D_{a^+}^{\alpha;\beta;\psi} f(x) = I_{a^+}^{\chi-\alpha;\psi} D_{a^+}^{\chi;\psi} f(x),$$

and

$${}^H D_{b^-}^{\alpha;\beta;\psi} f(x) = I_{b^-}^{\chi-\alpha;\psi} (-1)^n D_{b^-}^{\chi;\psi} f(x).$$

In what follows, we apply the above definitions to construct a new fractional operator with the sigmoid function as the kernel. Next it

can be built to model various nonlinear phenomena from different fields of applications in science.

3. New fractional operator and its models

3.1. New fractional operator

In the first place of this subsection, the Sigmoid function and Sigmoid type function are introduced as the kernel to construct a new fractional derivative operator [33–36]. The Sigmoid function which is also called logistic function in the field of statistics is a continuous non-linear function. It is usually asked to write digital logic [33], neural-mass models [34], image processing [35], optimized control and real-time speech [36].

The Sigmoid function [33] can be expressed as in the following form

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (9)$$

and a convenience of the Sigmoid function is its derivative (see Fig. 1)

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)). \quad (10)$$

From Fig. 1, we can see that the Sigmoid function have a symmetry point at (0,0.5), only half of the $x - \sigma(x)$ pairs can be computed by

$$\sigma(x)_{x>0} = 1 - \sigma(x)_{x<0} \quad \text{or} \quad \sigma(x)_{x<0} = 1 - \sigma(x)_{x>0}.$$

Taking $\psi(x) := \sigma(x) = \frac{1}{1+e^{-x}}$ into definition (2.3) as the kernel, we have

$$\begin{aligned} I_{a^+}^{\alpha, \sigma} f(x) &= \frac{1}{\Gamma(\alpha)} \int_a^x \frac{e^{-t}}{(1 + e^{-t})^2} \left(\frac{1}{1 + e^{-x}} - \frac{1}{1 + e^{-t}} \right)^{\alpha-1} f(t) dt \\ &= \frac{1}{\Gamma(\alpha)} \int_a^x \sigma(t)(1 - \sigma(t))(\sigma(x) - \sigma(t))^{\alpha-1} f(t) dt. \end{aligned} \quad (11)$$

Thus,

$$I_{a^+}^{(1-\beta)(n-\alpha), \sigma} f(x) = \frac{1}{\Gamma((1-\beta)(n-\alpha))} \int_a^x \sigma(t)(1 - \sigma(t))(\sigma(x) - \sigma(t))^{(1-\beta)(n-\alpha)-1} f(t) dt. \quad (12)$$

Next, according to the definition of the ψ -Hilfer fractional derivative, a new type fractional derivative operator can be derived as follows:

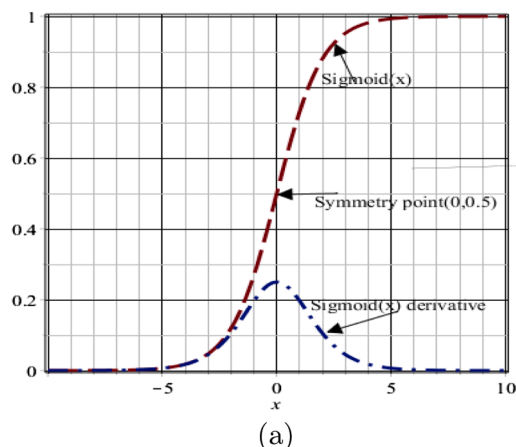


Fig. 1. The Sigmoid function and its derivative.

$$\begin{aligned}
{}^H D_{a^+}^{\alpha,\beta} f(x) &= I_{a^+}^{\beta(n-\alpha);\sigma} \left(\frac{1}{\psi'(x)} \frac{d}{dx} \right)^n I_{a^+}^{(1-\beta)(n-\alpha);\sigma} f(x) \\
&= \frac{1}{\Gamma(\beta(n-\alpha))} \int_a^x \sigma(t)(1-\sigma(t))(\sigma(x)-\sigma(t))^{\beta(n-\alpha)-1} F(t) dt \\
&= \frac{1}{\Gamma(\beta(n-\alpha))} \int_a^x \sigma(t)(1-\sigma(t))(\sigma(x)-\sigma(t))^{\beta(n-\alpha)-1} \cdot \left(\frac{1}{\sigma(t)(1-\sigma(t))} \frac{d}{dt} \right)^n \\
&\quad \cdot \frac{1}{\Gamma((1-\beta)(n-\alpha))} \int_a^t \sigma(t')(1-\sigma(t'))(\sigma(x)-\sigma(t'))^{(1-\beta)(n-\alpha)} f(t') dt' dt'.
\end{aligned} \tag{13}$$

In particular, we taking the following special cases:

Case 1. When $\beta \rightarrow 0, n = 1$, Eq. (13) becomes

$${}^H D_{a^+}^{\alpha} f(x) = \frac{1}{\Gamma(1-\alpha)} \int_a^x \frac{1}{\sigma(x)-\sigma(t)} \frac{d}{dt} \int_a^t \sigma(t')(1-\sigma(t'))(\sigma(x)-\sigma(t'))^{1-\alpha} f(t') dt' dt'. \tag{14}$$

Case 2. When $\beta \rightarrow 1, n = 1$, Eq. (13) yields

$${}^H D_{a^+}^{\alpha,\beta} f(x) = \frac{1}{\Gamma(1-\alpha)} \int_a^x \frac{1}{(\sigma(x)-\sigma(t))^\alpha} \frac{d}{dt} \int_a^t \sigma(t')(1-\sigma(t')) f(t') dt' dt'. \tag{15}$$

Case 3. When $\beta \rightarrow 0$, Eq. (13) has

$${}^H D_{a^+}^{\alpha} f(x) = \frac{1}{\Gamma(n-\alpha)} \int_a^x (\sigma(t)(1-\sigma(t)))^{1-n} \left(\frac{d}{dt} \right)^n \int_a^t \sigma(t')(1-\sigma(t'))(\sigma(x)-\sigma(t'))^{n-\alpha} f(t') dt' dt'. \tag{16}$$

Case 4. When $\beta \rightarrow 1$, Eq. (13) shows

$${}^H D_{a^+}^{\alpha,\beta} f(x) = \frac{1}{\Gamma(n-\alpha)} \int_a^x \sigma(t)(1-\sigma(t))^{1-n} (\sigma(x)-\sigma(t))^{n-\alpha-1} \left(\frac{d}{dt} \right)^n \int_a^t \sigma(t')(1-\sigma(t')) f(t') dt' dt'. \tag{17}$$

From Eqs. (14)–(17), we can find that the Eq. (14) and (15) are equivalent with the condition $\beta \rightarrow 0, n = 1$ and $\beta \rightarrow 1, n = 1$, respectively. In a similar way, the Eq. (16) and (17) are equivalent with the condition $\beta \rightarrow 0$ and $\beta \rightarrow 1$, respectively.

Property 1. The new type fractional derivative (13) is bounded operator for all $n-1 < \alpha < n$ and $0 \leq \beta \leq 1$, given by

$$\| {}^H D_{0^+}^{\alpha,\beta} f \|_{C_{r,\sigma}} \leq K \| f \|_{C_{r,\sigma}^n}, \tag{18}$$

where

$$K = \left(\frac{1-e^{-b}}{2(1+e^{-b})} \right)^{n-\alpha} \frac{1}{\Gamma(n-\gamma+\alpha)\Gamma(\gamma-\alpha+1)}.$$

and σ is Sigmoid function (9).

Proof. A similar proof process is shown in Refs. [23,25]. $\square$

Property 2. If considering the function f is power function on Sigmoid function, then we have

$${}^H D_{a^+}^{\alpha,\beta} f(x) = \frac{\Gamma(\delta)}{\Gamma(\delta-\alpha)} \left(\frac{1}{1+e^{-x}} - \frac{1}{1+e^{-a}} \right)^{\delta-\alpha-1}, \quad \delta > n, \tag{19}$$

where $n-1 < \alpha < n$ and $0 \leq \beta \leq 1$.

Proof. At the beginning of, we assume that the function f can be expressed as

$$f(x) = (\sigma(x) - \sigma(a))^{\delta-1} = \left(\frac{1}{1+e^{-x}} - \frac{1}{1+e^{-a}} \right)^{\delta-1}, \delta > n. \quad (20)$$

By taking Lemma 1 and Lemma 2, we get

$$\begin{aligned} {}^H D_{a^+}^{\alpha, \beta} f(x) &= I_{a^+}^{\gamma-\alpha; \sigma} D_{a^+}^{\gamma; \sigma} f(x) \\ &= I_{a^+}^{\gamma-\alpha; \sigma} D_{a^+}^{\gamma; \sigma} (\sigma(x) - \sigma(a))^{\delta-1} \\ &= I_{a^+}^{\gamma-\alpha; \sigma} D_{a^+}^{\gamma; \sigma} \left(\frac{1}{1+e^{-x}} - \frac{1}{1+e^{-a}} \right)^{\delta-1} \\ &= I_{a^+}^{\gamma-\alpha; \sigma} \left[\frac{\Gamma(\alpha)}{\Gamma(\delta-\gamma)} \left(\frac{1}{1+e^{-x}} - \frac{1}{1+e^{-a}} \right)^{\delta-\gamma-1} \right] \\ &= \frac{\Gamma(\alpha)}{\Gamma(\delta-\gamma)} \left(\frac{1}{1+e^{-x}} - \frac{1}{1+e^{-a}} \right)^{\delta-\alpha-1}. \end{aligned} \quad (21)$$

This property was completed. $\square$

Remark 2. In order to control the Sigmoid function slope and better applications of natural science, a parameter α is proposed [36]. Here the Sigmoid function is called the Sigmoid type function. That is

$$\sigma(x) = \frac{1}{1+e^{-\alpha x}}, \quad (22)$$

and its derivative

$$\sigma'(x) = \alpha \sigma(x)(1 - \sigma(x)). \quad (23)$$

(see Figs. 2, 3) In a similar matter, a new fractional derivative operator with the Sigmoid type function as the kernel can be given by

$$\begin{aligned} {}^H D_{a^+}^{\alpha, \beta} f(x) &= \frac{1}{\Gamma(\beta(n-\alpha))} \frac{1}{\alpha^{n-2}} \int_a^x \sigma(t)(1-\sigma(t))(\sigma(x)-\sigma(t))^{\beta(n-\alpha)-1} \cdot \left(\frac{1}{\sigma(t)(1-\sigma(t))} \frac{d}{dt} \right)^n \\ &\quad \cdot \frac{1}{\Gamma((1-\beta)(n-\alpha))} \int_a^t \sigma(t')(1-\sigma(t'))(\sigma(x)-\sigma(t'))^{(1-\beta)(n-\alpha)} f(t') dt' dt'. \end{aligned} \quad (24)$$

3.2. Models

In view of the nonlinear phenomena in nature, one may apply the fractional derivative to express the mathematical models. Now we give examples and open problems for the population growth, shallow water wave and reaction-diffusion process, etc.

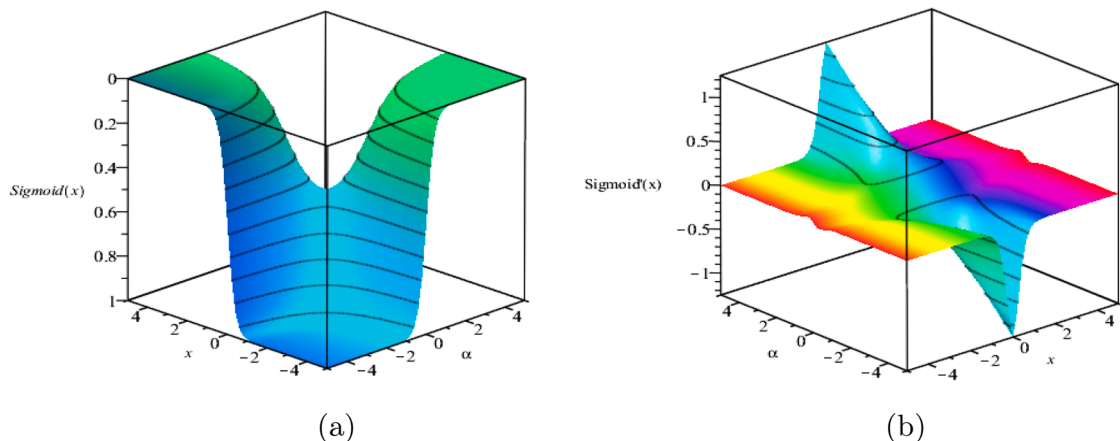


Fig. 2. The parameter value α and the spatial variable x effect the Sigmoid function (22) and its derivative (23).

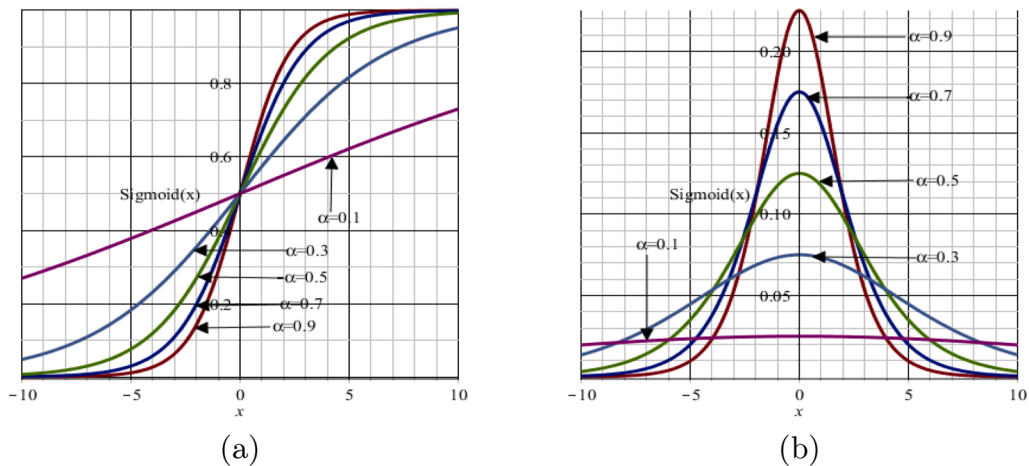


Fig. 3. The parameter value α for the Sigmoid function (22) and its derivative (23) yield different affects.

Model 1. The mathematical model of the population growth [6] within the fractional derivative (13) can be given as

$$({}^H D_{a^+}^{\alpha,\beta} f)(t) = s_0 \exp(\lambda t), \quad (25)$$

where s_0 is the initial population size and λ is the population growth rate.

Model 2. The mathematical models of the shallow water wave [37–39] within the fractional derivative (13) can be expressed by (I) the Korteweg-de Vries equation [37]

$$({}^H D_{a^+}^{\alpha,\beta} f)(x, t) + f_{xxx} - 6ff_x = 0, \quad (26)$$

(II) the Burger's equation [38]

$$({}^H D_{a^+}^{\alpha,\beta} f)(x, t) - \varepsilon f_{xx} + ff_x = 0, \quad (27)$$

where ε is a parameter, related to the Reynolds number $R = 1/\varepsilon$ in the viscous fluid flow problem.

(III) the general nonlinear shallow-water wave equations [39]

$$\begin{cases} ({}^H D_{a^+}^{\alpha,\beta} f_1)(x, y, t) + f_1 f_{1x} + f_2 f_{1y} - af_2 + bf_{3x} = 0, \\ ({}^H D_{a^+}^{\alpha,\beta} f_2)(x, y, t) + f_1 f_{2x} + f_2 f_{2y} + af_1 + bf_{3y} = 0, \\ ({}^H D_{a^+}^{\alpha,\beta} f_3)(x, y, t) + [f_1(D + f_3)]_x + [f_2(D + f_3)]_y = 0, \end{cases} \quad (28)$$

where f -the earth's rotation and g -the acceleration of gravity, are the Coriolis parameters. The D is positive below the equilibrium level.

Model 3. The mathematical models of the reaction-diffusion process [40,41] within the fractional derivative (13) can be written as (I) the one dimensional reaction-diffusion equation [40]

$$({}^H D_{a^+}^{\alpha,\beta} f)(x, t) = f_{xx} + u(f), \text{ with } u(0) = u(1) = 1, \quad (29)$$

(II) the reaction-diffusion equation [41]

$$({}^H D_{a^+}^{\alpha,\beta} f)(x, t) = (f^m)_{xx} + \lambda f^n, \{x, t\} \in R \times (0, +\infty), \quad (30)$$

where $m > 1, \lambda > 0$ and $n \in R$.

4. Applications

In this section, we give two simple examples, namely the population growth model and the oscillatory model, in the sense of the new fractional derivative.

Example 1. The population growth model was proposed by the English economist Thomas Malthus [6]. That is to say,

$$f'(t) = \lambda f(t), \quad (31)$$

where λ is the population rate.

It has the following form solution

$$f(t) = s_0 \exp(\lambda t),$$

where $f(0) = s_0$ is the initial population size. Here we mainly focus on the above model within the new fractional derivative operator, which reads

$$({}^H D_{a^+}^{\alpha, \beta} f)(t) = s_0 \exp(\lambda t), \quad (32)$$

where $f(0) = s_0$ is the initial population size and λ is the population growth rate.

After rigorous calculations and along with property 2, we can get the solution to Eq. (32)

$$f(t) = s_0 \sum_{k=0}^{\infty} \frac{\lambda^k}{\Gamma(k\mu - \chi + 1)} \left(\frac{1}{1 + e^{-t}} - \frac{1}{1 + e^{-a}} \right)^{k\mu - \chi}, \quad (33)$$

where

$$\mu = n(1 - \beta) + \alpha\beta, \chi = (1 - \beta)(n - \alpha)$$

In particular, if take $\lambda > 0, n - 1 < \alpha < n$ and $0 \leq \beta \leq 1$, according to Lemma 6 [25], we have

$$f(t) = s_0 E_{\alpha} \left(\lambda \left(\frac{1}{1 + e^{-t}} - \frac{1}{1 + e^{-a}} \right)^{\alpha} \right), \quad (34)$$

where $E_{\alpha}(\cdot)$ is a one parameter Mittag-Leffler function.

Example 2. The oscillatory model in the sense of the new fractional derivative of the form [42,43]

$$\begin{cases} {}^H D_{a^+}^{\alpha, \beta} x(t) + f_1(t, x) = \omega(t) + f_2(t, x), \\ I_{a^+}^{1-\gamma, \sigma} x(t) = b_1, \quad b_1 > 0, \end{cases} \quad (35)$$

where f_1, f_2, ω are continuous functions, and $I_{a^+}^{1-\gamma, \sigma}$ is ψ RL fractional integral [2], with

$$0 < \alpha < 1, 0 \leq \beta < 1, \gamma = \alpha + \beta(1 - \alpha)$$

The solution of system (35) via property 2 and integral transformation on both sides in the form, was obtained

$$\begin{aligned} x(t) = & \frac{b_1 \left(\frac{1}{1 + e^{-t}} - \frac{1}{1 + e^{-a}} \right)^{\gamma-1}}{\Gamma(\gamma)} \\ & + \Gamma(\alpha) \int_a^t \frac{e^{-s}}{(1 + e^{-s})^2} \left(\left(\frac{1}{1 + e^{-t}} - \frac{1}{1 + e^{-s}} \right)^{\alpha-1} [\omega(s) + f_2(s, x(s)) - f_1(s, x(s))] \right) ds. \end{aligned} \quad (36)$$

Remark 3. If the solution (35) has arbitrarily large zeros on $(0, \infty)$, then it is called oscillatory; otherwise, it is called non-oscillatory.

5. Conclusions and discussions

In this article, a new fractional derivative operator with the help of the Sigmoid function as the kernel, was proposed. Then some new mathematical models, such as the population growth, the shallow water wave and reaction-diffusion process within the fractional derivative, were also given. We hope that these new models can show more evolutionary mechanisms of these phenomena. Meanwhile, we listed a few open problems on this topic. For example, (i) we can take the Sigmoid type function [36]

$$\sigma(x) = \frac{1}{1 + e^{g(x)}},$$

as the kernel to derive more general fractional derivative operator, where $g(x)$ is a smooth function.

(ii) We can develop new numerical methods to deal with these proposed models.

For these problems of the above states, we will consider them in the near future.

Declaration of Competing Interest

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication.

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